

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11.	(a)			Diagram should show: Polarisation of N—H or O—H bonds in 2-aminoethanol (1) Lone pairs of electrons used in hydrogen bonding to nitrogen or oxygen atoms (1) Polarisation of water molecules (1) Hydrogen bonds indicated between 2-aminoethanol and water molecules using relevant nitrogen / oxygen and hydrogen atoms (1) If no water then 3 marks maximum	2	2		4		
	(b)			The forces of attraction between molecules of 2-aminoethanol are stronger than the attractive forces between molecules of 1,2-diaminoethane (as the former has a higher boiling temperature) (1) This suggests that intermolecular hydrogen bonding between / involving O and H is stronger than the hydrogen bonding between N and H (1) This is (probably) due to a greater electronegativity difference between O and H than between N and H / O more electronegative than N / size considerations (1)		1	2	3		

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11.	(c)			Indicative content 1 aqueous sulfuric acid in burette (accept aqueous ammonia in burette) 2 measure volume of ammonia into flask 3 use of an indicator (not universal indicator) 4 titrate with aqueous sulfuric acid until colour of indicator just changes 5 read burette and repeat without indicator / use of decolorising charcoal and filter 6 concentrate neutralised solution 7 cool (concentrated) solution / leave to evaporate over time 8 filter and dry crystals	3	3		6		6
				5-6 marks The method provided leads to pure dry crystals of ammonium sulfate. <i>The candidate constructs a relevant and logically structured account including all key elements of the indicative content. Scientific conventions and vocabulary are used accurately throughout.</i> 3-4 marks The method outlined leads to the production of a solution of ammonium sulfate. <i>The candidate constructs a logically structured account including the main elements of the indicative content. The use of scientific conventions and vocabulary are generally sound.</i> 1-2 marks The method provided leads to the production of a solution that contains ammonium sulfate. <i>The candidate has given an outline method of the production of ammonium sulfate but a number of key points are missing. There is some evidence of the correct use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give an answer worthy of credit.</i>						
				Question 11 total	5	6	2	13	0	6